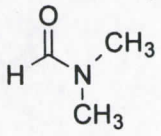
Organic Cume 09-10-2015

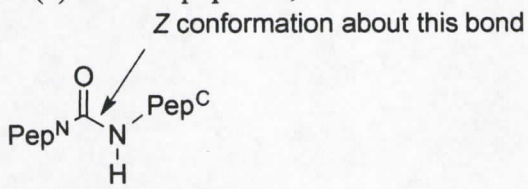
1. (a) The two methyl groups in dimethylformamide (DMF) have different chemical shifts in its proton spectrum. Explain why.



DMF

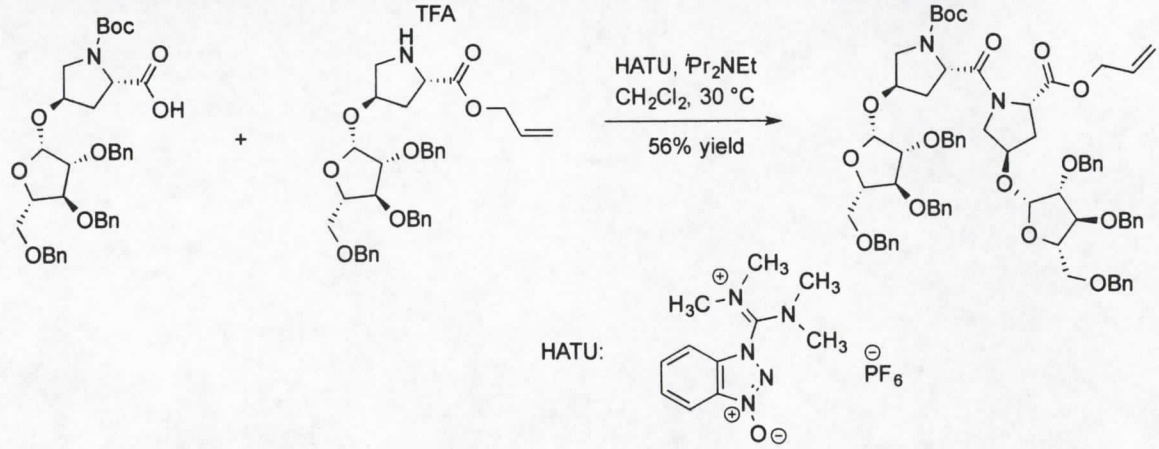
(b) What kind of NMR experiment can you use to probe your reasoning for question (a)?

2. (a) In most peptides, the amide bonds have the *Z* conformation; explain why.

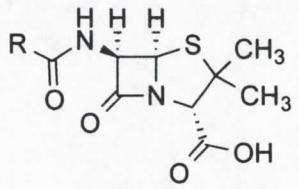


(b) One particular amino acid residue in the pep^C position adopts the *E* conformation in some cases. Which amino acid residue should be most likely to assume an *E* conformation, and why?

3. Write an electron-pushing mechanism for the following reaction discussed by Professor Carol M. Taylor of LSU in her lecture.



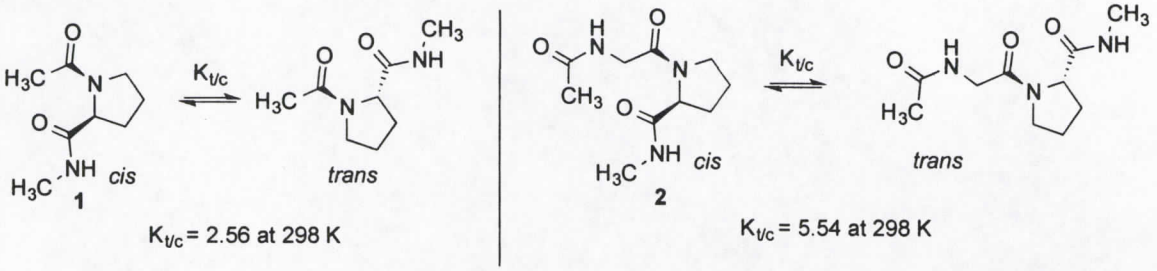
4. Amides are generally stable towards hydrolysis. But one of the two amide bonds in penicillin shown blow is rapidly hydrolyzed by the acidic environment of the stomach. Draw out the structure of the hydrolysis product and also explain the chemoselectivity for the hydrolysis.



Penicillin

5. Professor Carol M. Taylor of LSU has extensively studied factors affecting conformation in Proline-containing peptides. Give brief explanations of the two phenomena reported by her group shown below.

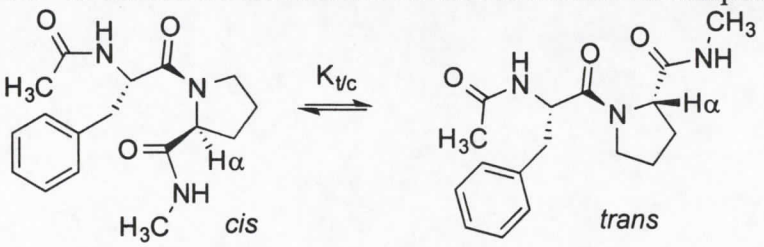
(a) Change of the *N*-terminal residue from Ac- (**1**) to Ac-Gly- (**2**) increases the population of the *trans* conformer.



$K_{t/c}$ = 2.56 at 298 K

$K_{t/c}$ = 5.54 at 298 K

(b) Replacing Gly of **2** with Phe (**3**), $K_{t/c}$ shows a decrease from 5.54 to 2.56 at 298 K. The ^{1}H chemical shifts for Hα of the Pro residue in compounds **2** and **3** are listed below.



3 $K_{t/c}$ = 2.56 at 298 K

^{1}H NMR (D_2O, 298 K)

	2	**3**
δHα (trans)	4.23	4.38
δHα (cis)	4.41	3.47